TASK 3: Attach and Detach Disk from Windows VM

Objective

To attach a new data disk to an existing Windows Virtual Machine, initialize and format the disk, assign a drive letter, and then detach the disk from the VM.

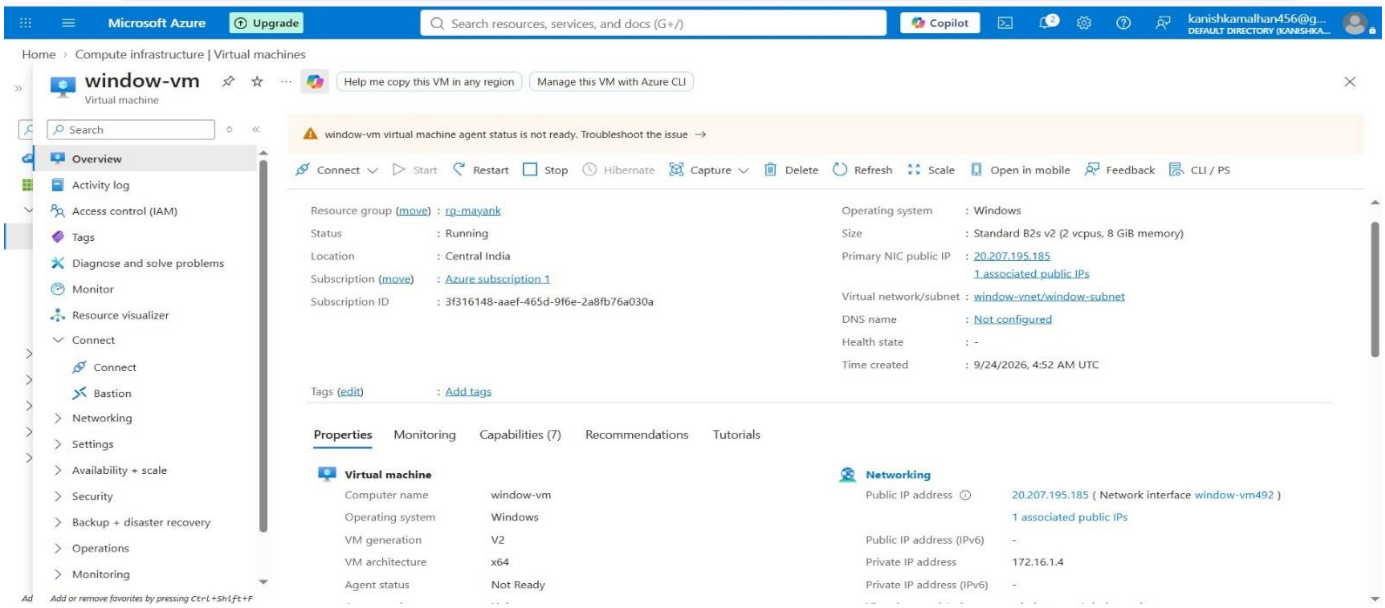
Theory

An Azure Data Disk is an additional managed disk that can be attached to a Virtual Machine to increase storage capacity. Data disks are commonly used to store application data, databases, logs, and user files separately from the operating system disk. Azure allows administrators to attach, initialize, use, and detach disks as required without affecting the operating system.

Procedure

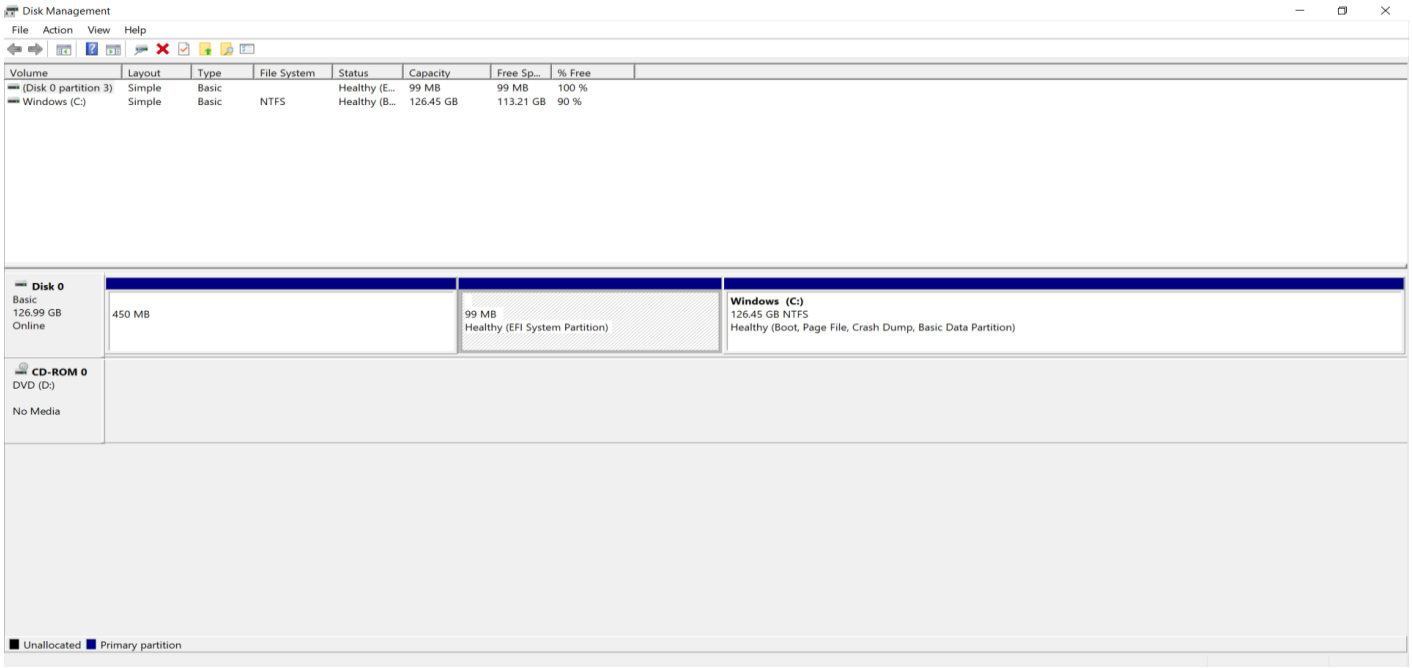
Step 1: Open Virtual Machine

Navigate to **Azure Portal** → **Virtual Machines** and select the target Windows VM.



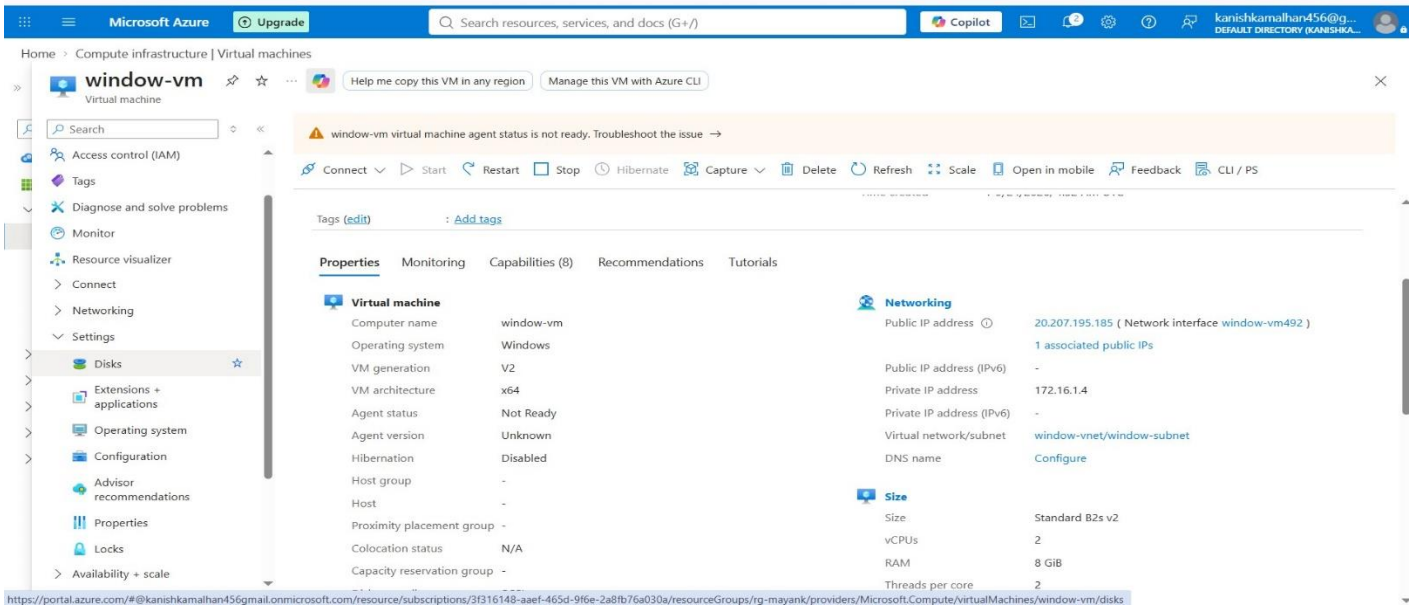
Step 2: Connect to the Windows VM

Connect to the VM using Remote Desktop Protocol (RDP).



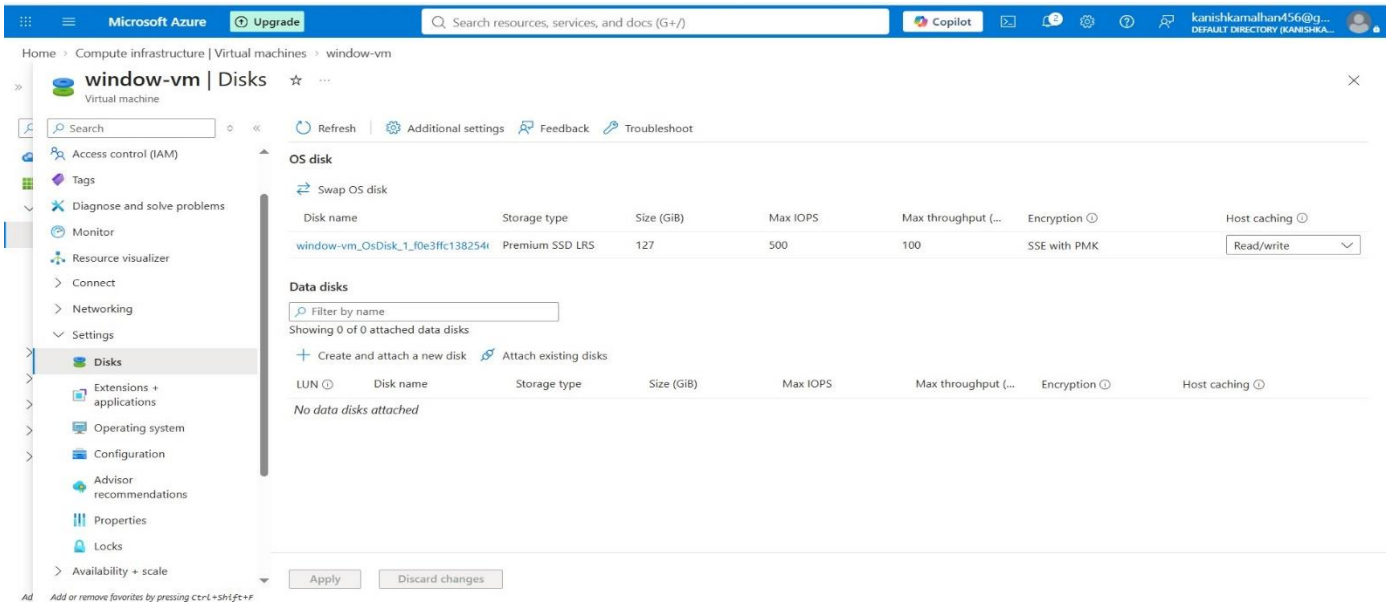
Step 3: Open Disks Blade

Disks Configuration Page , From the left menu, click **Disks**.



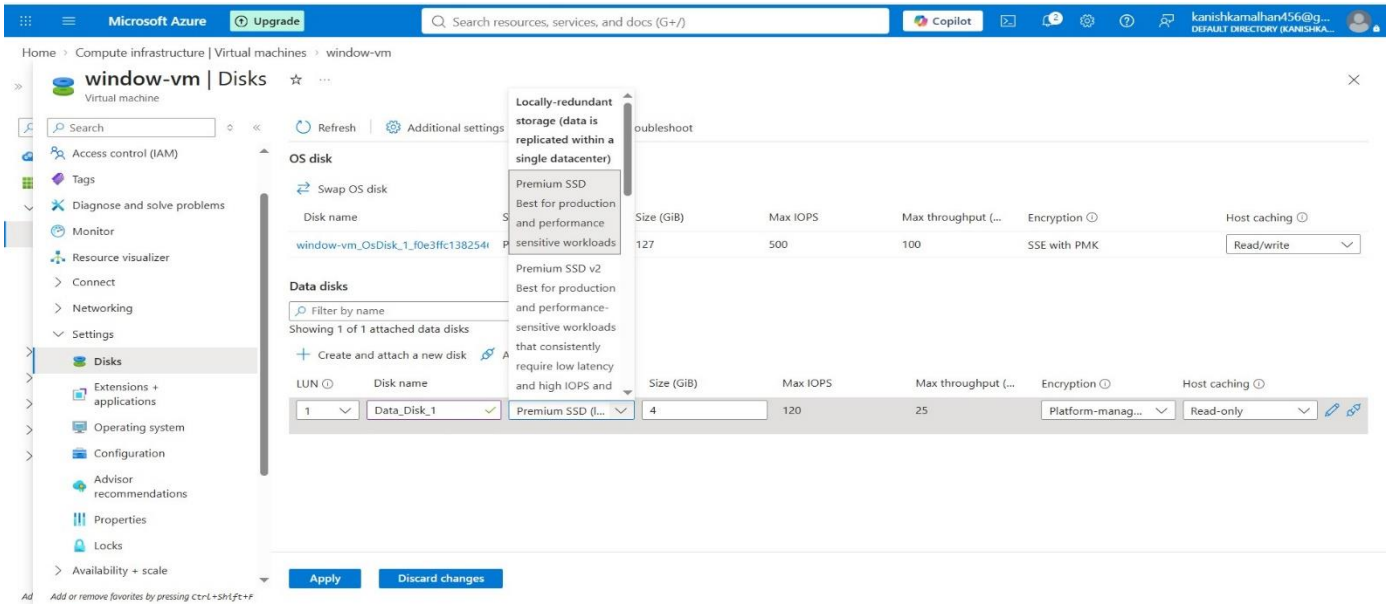
Step 4: Create and Attach New Disk

Click **Create and attach a new disk**.



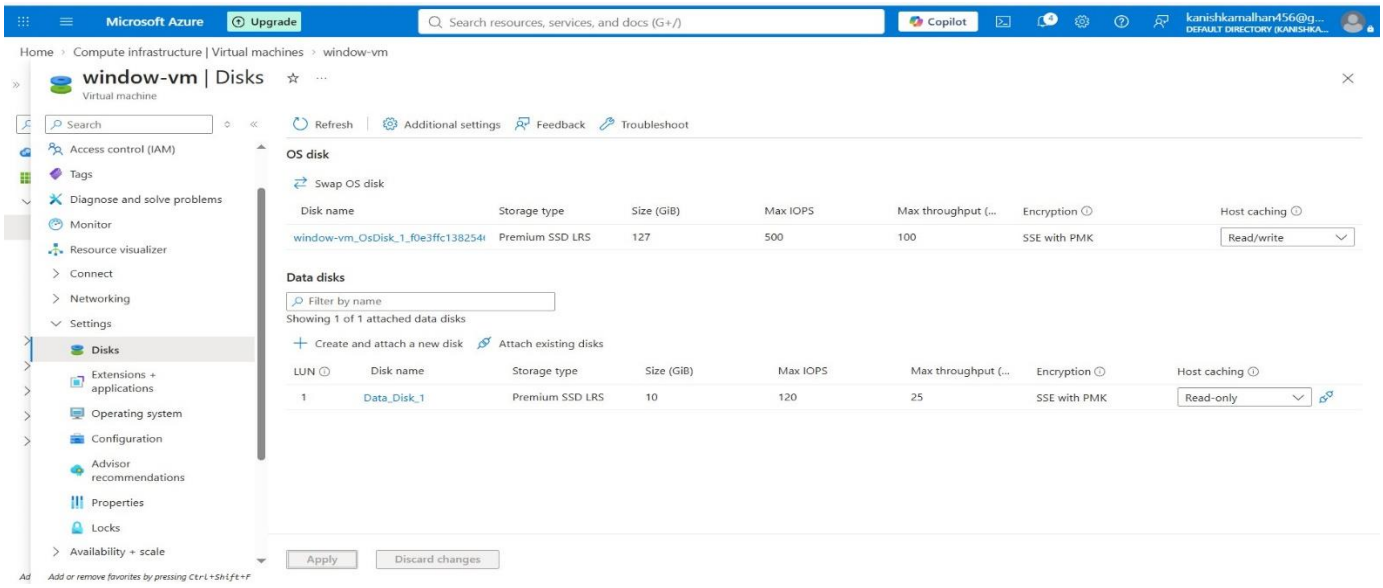
Step 5: Configure Disk Details

Enter the disk name and select the required disk size.



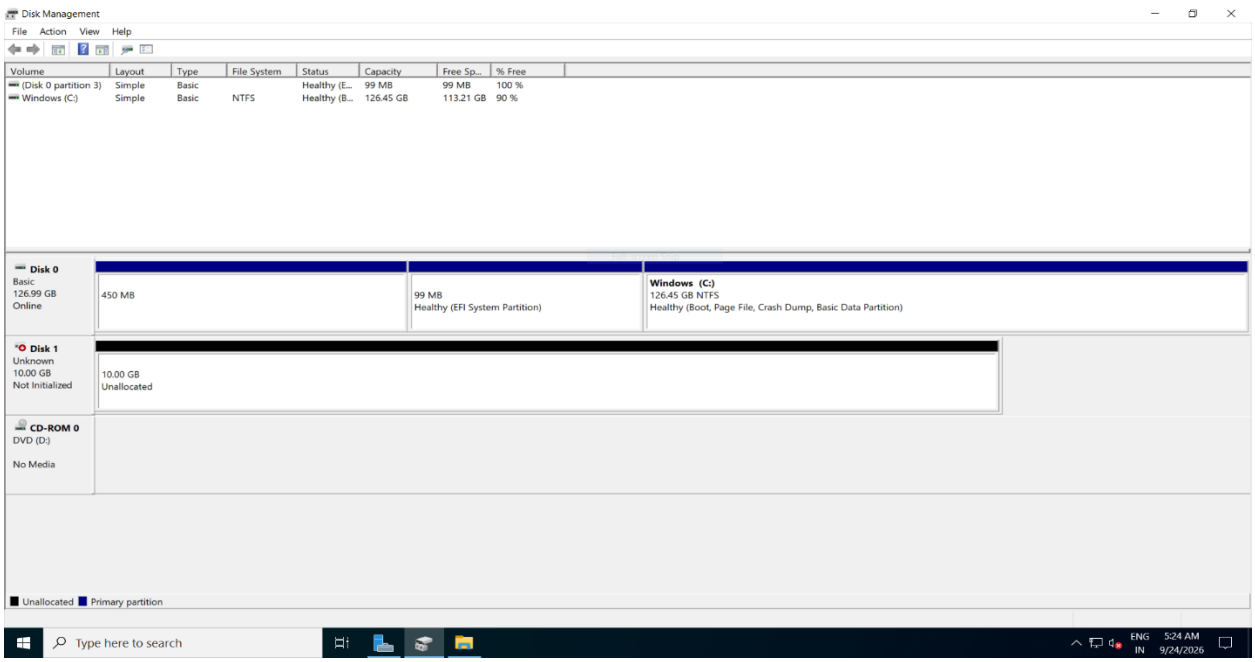
Step 6: Attach the Disk

Click **Apply/Save** to attach the new disk to the VM. Disk Attached Successfully



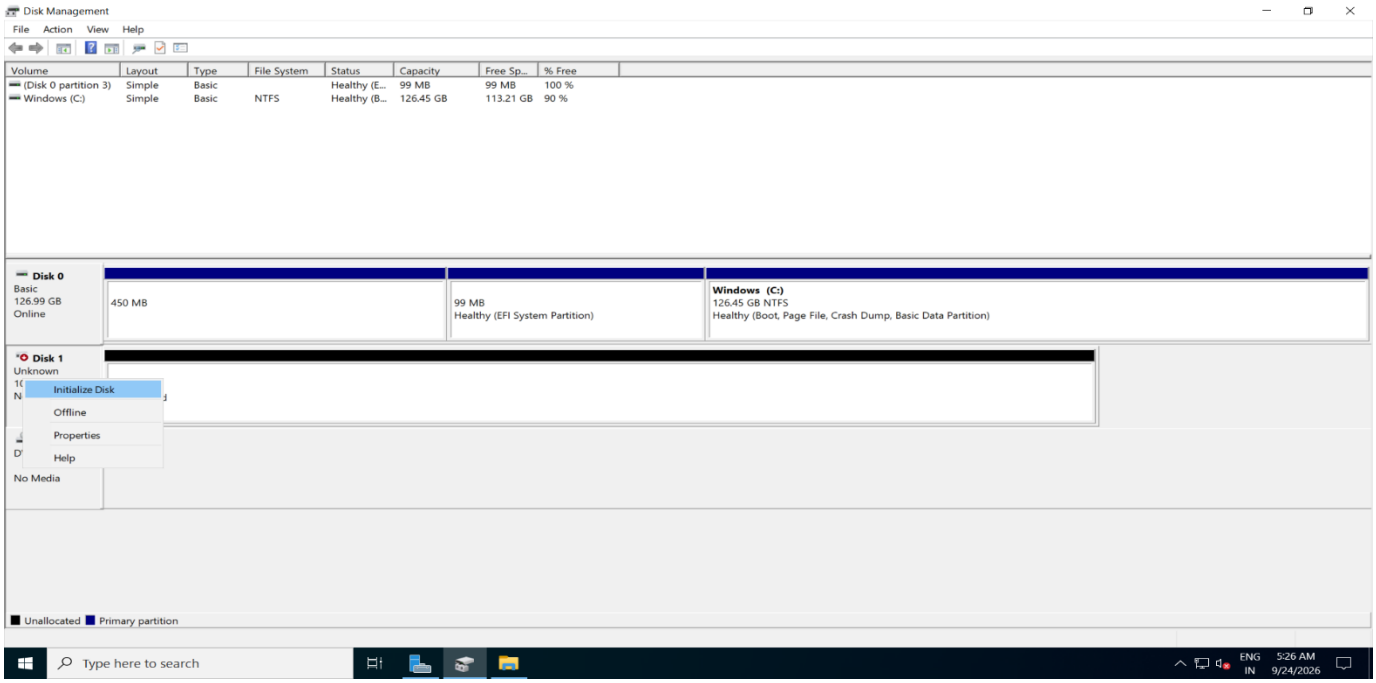
Step 7: Open Disk Management

Open **Disk Management** from Server Manager or Run → `diskmgmt.msc`. Disk Management Showing Newly Attached Disk



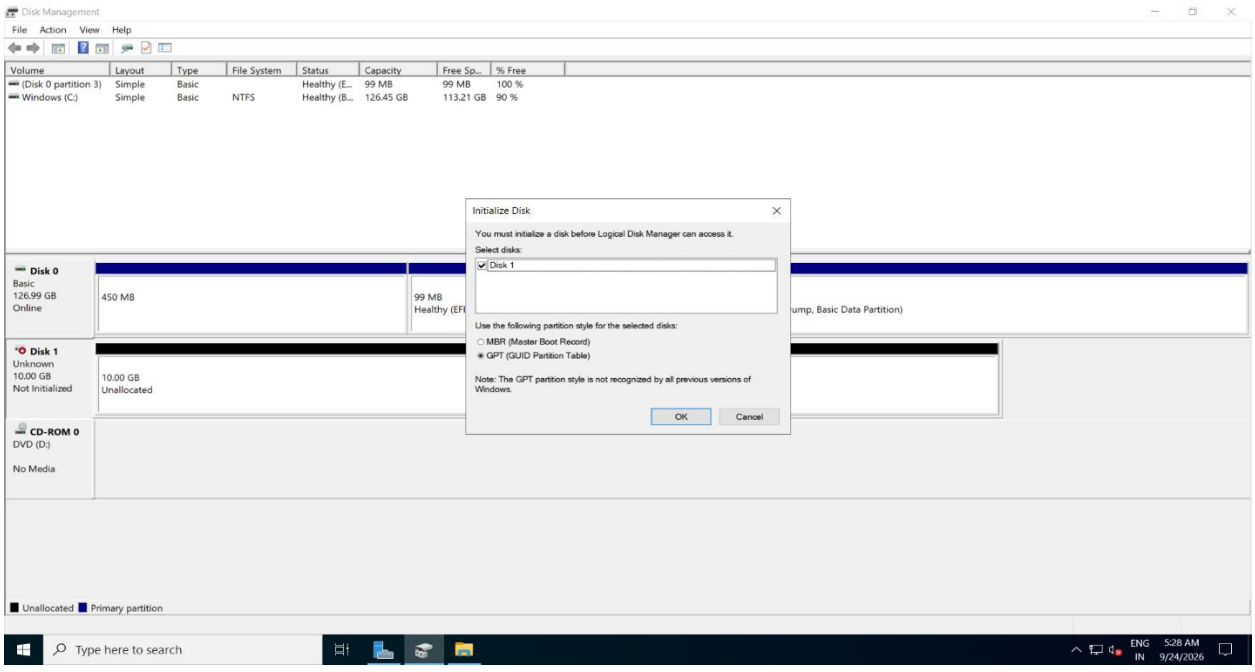
Step 8: Initialize the Disk

Right-click the new disk and select **Initialize Disk**. Choose **GPT (GUID Partition Table)**.



Step 9: Create New Simple Volume

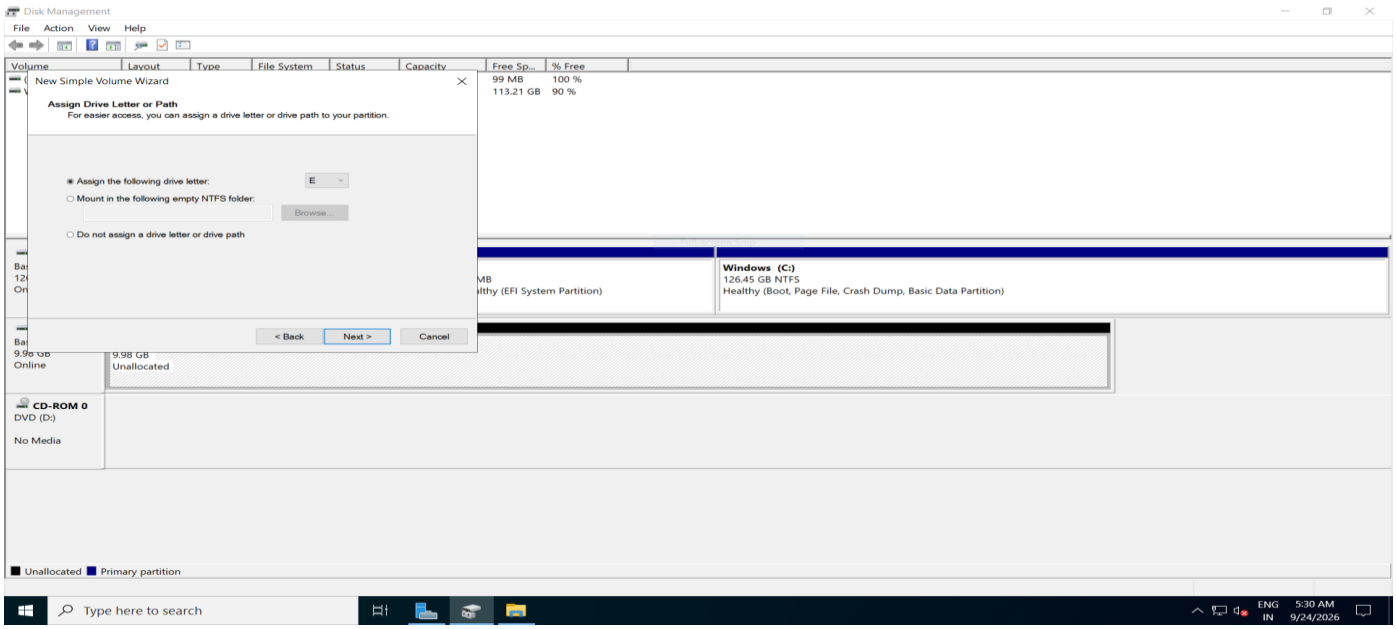
Right-click the unallocated space and select **New Simple Volume**.



Step 10: Assign Drive Letter and Mount the Volume

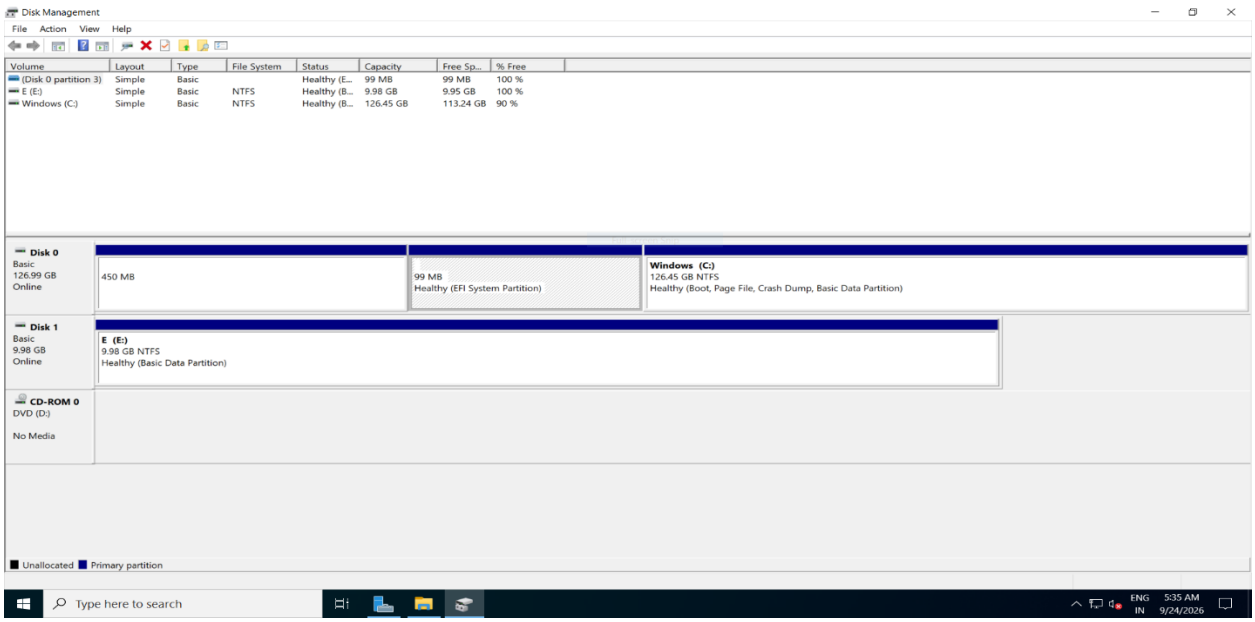
Assign the drive letter (**E:**) to the new volume.

Note: After assigning the drive letter, Windows automatically mounts the volume and makes it accessible through File Explorer.



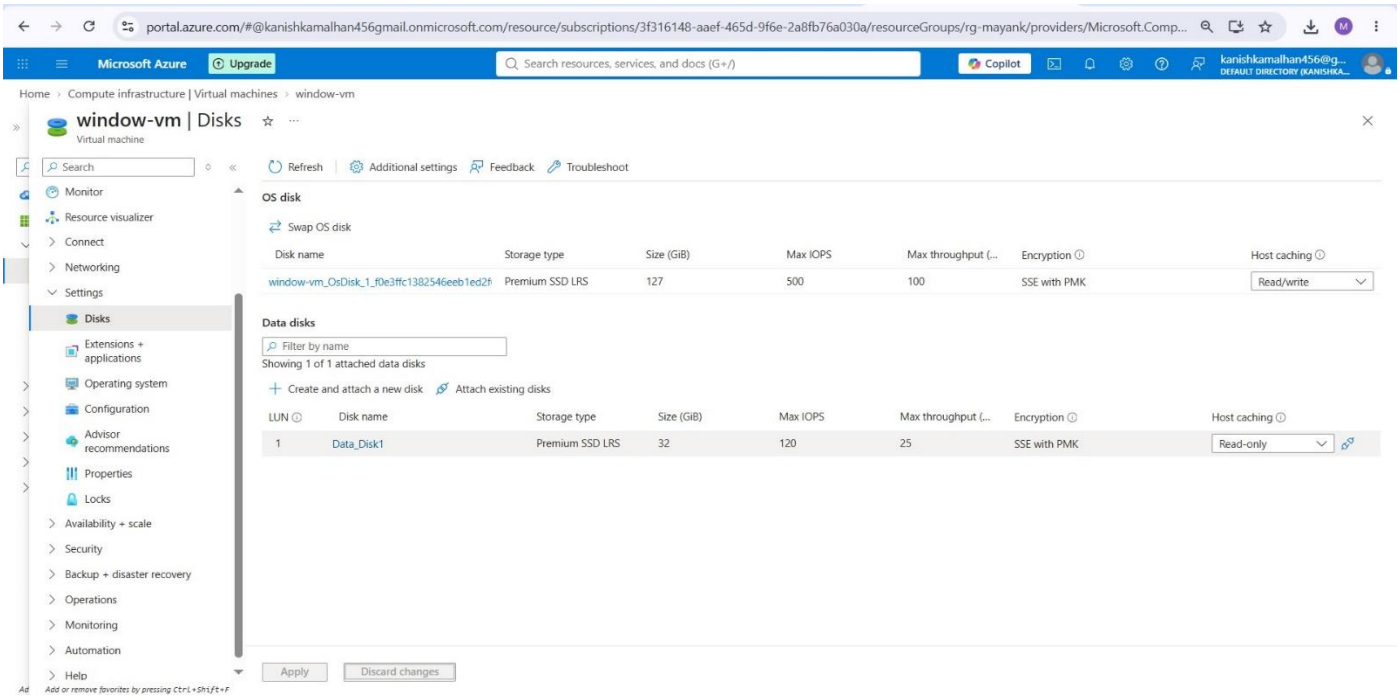
Step 11: Verify Disk Availability

Confirm that the new E: drive is visible and healthy. Attached Disk (E:) Visible in Disk Management.

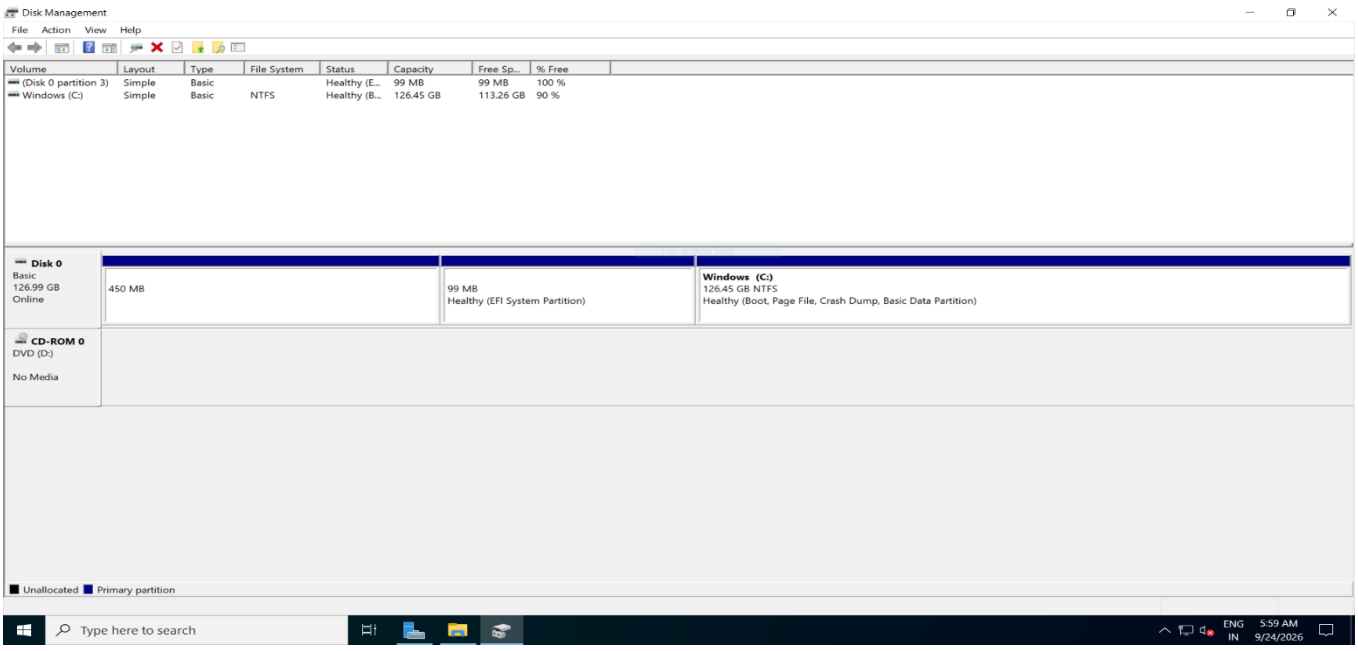


Step 12: Detach the Disk

Navigate to **Azure Portal** → **VM** → **Disks**, remove the data disk, and save the configuration.



Step 12: Verify Disk is Detached



Observation

- Successfully attached a new Azure Managed Data Disk to the Windows Virtual Machine.
- Verified that the newly attached disk was visible in **Disk Management**.
- Initialized the disk using the **GPT (GUID Partition Table)** partition style.
- Created a **New Simple Volume** on the attached disk.
- Formatted the volume using the **NTFS** file system.
- Assigned the drive letter **E:** to the volume.
- Verified that the disk was automatically mounted and accessible through **File Explorer/Disk Management**.
- Confirmed that the disk was available for data storage and usage.
- Successfully detached the data disk from the virtual machine through the Azure Portal.
- Verified that the detached disk was no longer associated with the Windows Virtual Machine.

Result

The Azure Managed Data Disk was successfully attached to the Windows Virtual Machine, initialized, formatted, mounted using drive letter **E:**, verified for use, and detached successfully from the Virtual Machine.

Attach and Detach a Data Disk to a Linux Virtual Machine

Objective

To attach an Azure Managed Data Disk to a Linux Virtual Machine, verify the disk within the Linux operating system, create a file system, mount the disk for use, validate read/write operations, and safely detach the disk from the virtual machine.

Theory

Azure Managed Disks are block-level storage volumes managed by Microsoft Azure and used with Virtual Machines. A Data Disk provides additional storage capacity beyond the operating system disk and can be attached or detached as per business requirements.

In Linux, a newly attached disk must be formatted with a file system and mounted to a directory before it can be used for storing data. Before detaching a disk, it should be unmounted to prevent data corruption and ensure safe removal.

Components Used

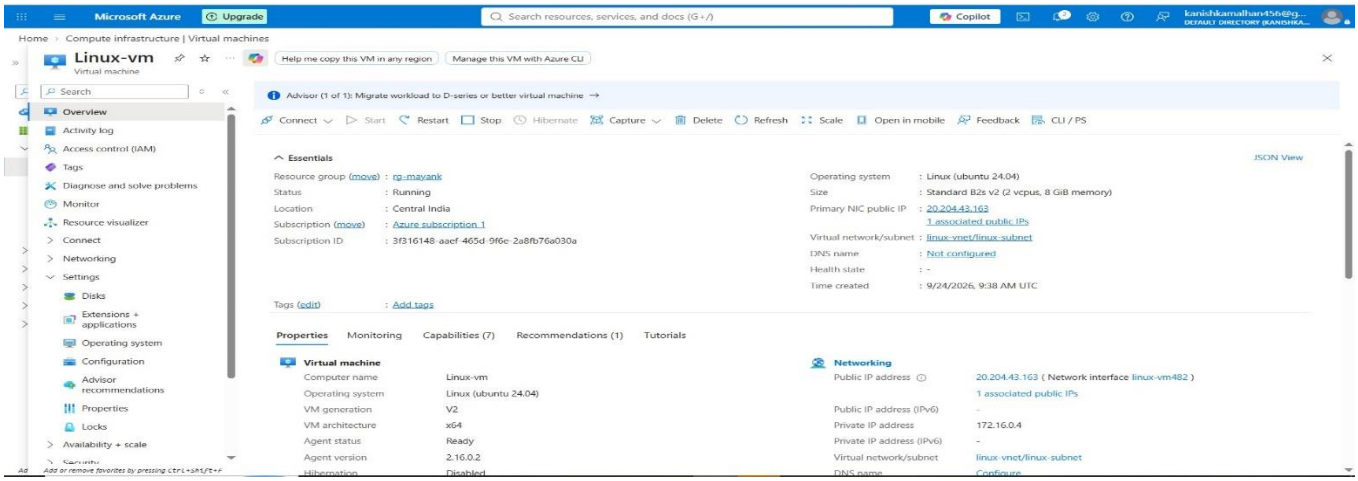
- Microsoft Azure Portal

- Azure Linux Virtual Machine (Linux-vm)
- Azure Managed Data Disk
- Ubuntu 24.04 Operating System
- SSH Terminal

Procedure

Step 1: Open the Linux Virtual Machine

1. Sign in to the Azure Portal.
2. Navigate to **Virtual Machines**.
3. Select the virtual machine named **Linux-vm**.
4. Verify that the VM is running successfully.

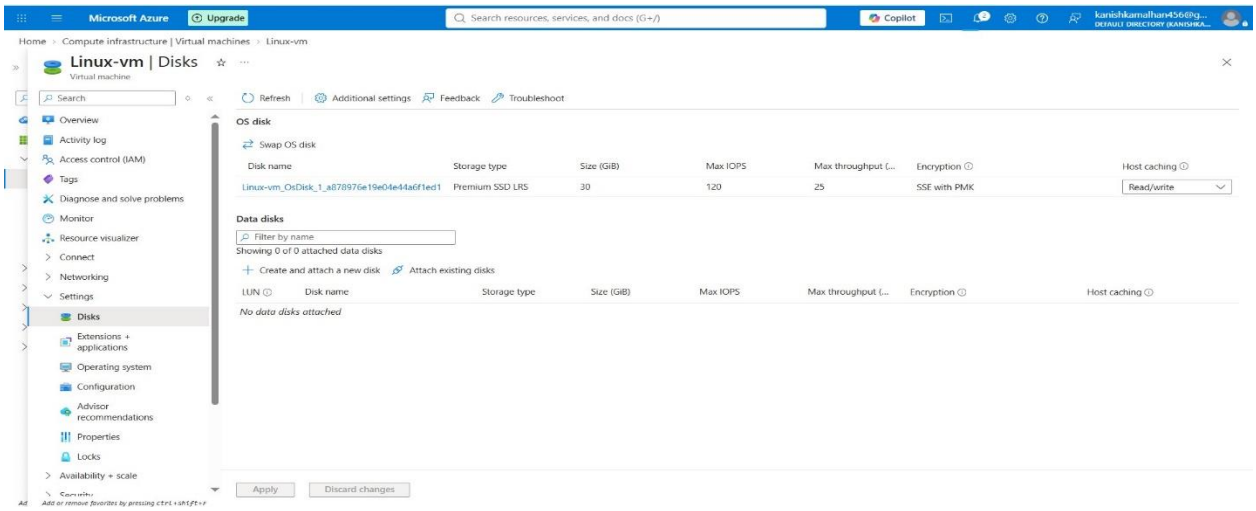


Step 2: Open Disk Settings

1. From the left navigation pane, under **Settings**, click **Disks**.
2. Review the currently attached disks.
3. Verify that only the OS disk is attached.

Existing Disk Details

- OS Disk: Linux-vm_OsDisk
- Storage Type: Premium SSD LRS
- Disk Size: 30 GiB

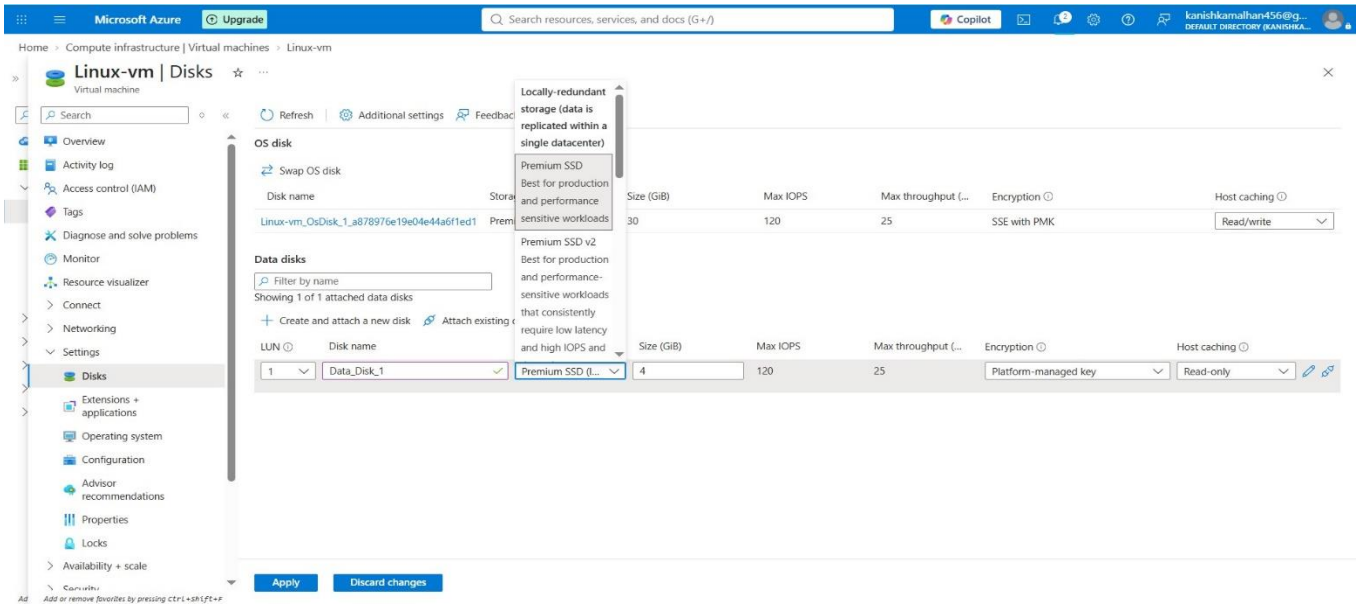


Step 3: Create and Attach a New Data Disk

1. Under **Data Disks**, click **Create and attach a new disk**.
2. Enter the disk name as **Data_Disk_1**.
3. Select **Premium SSD LRS** as the storage type.
4. Specify the desired disk size.
5. Click **Apply**.

New Disk Details

- Disk Name: Data_Disk_1
- Storage Type: Premium SSD LRS
- Encryption: Platform Managed Key

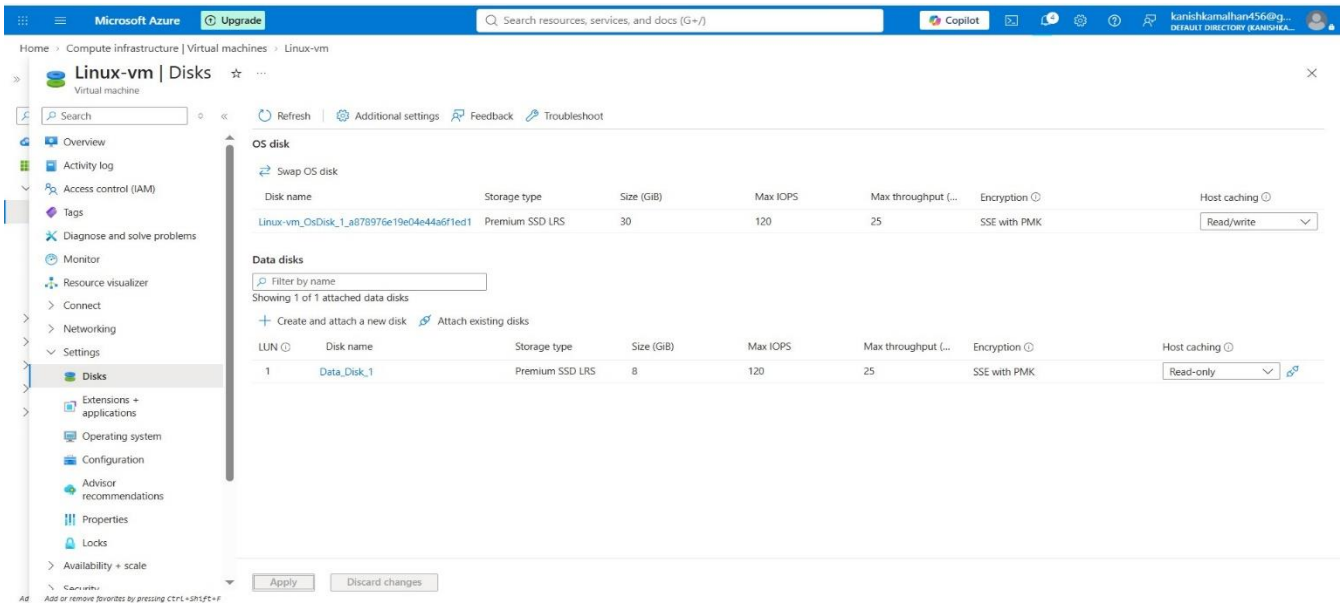


Step 4: Verify Disk Attachment

1. Refresh the Disks page.
2. Confirm that **Data_Disk_1** appears under the Data Disks section.
3. Verify that the disk is successfully attached.

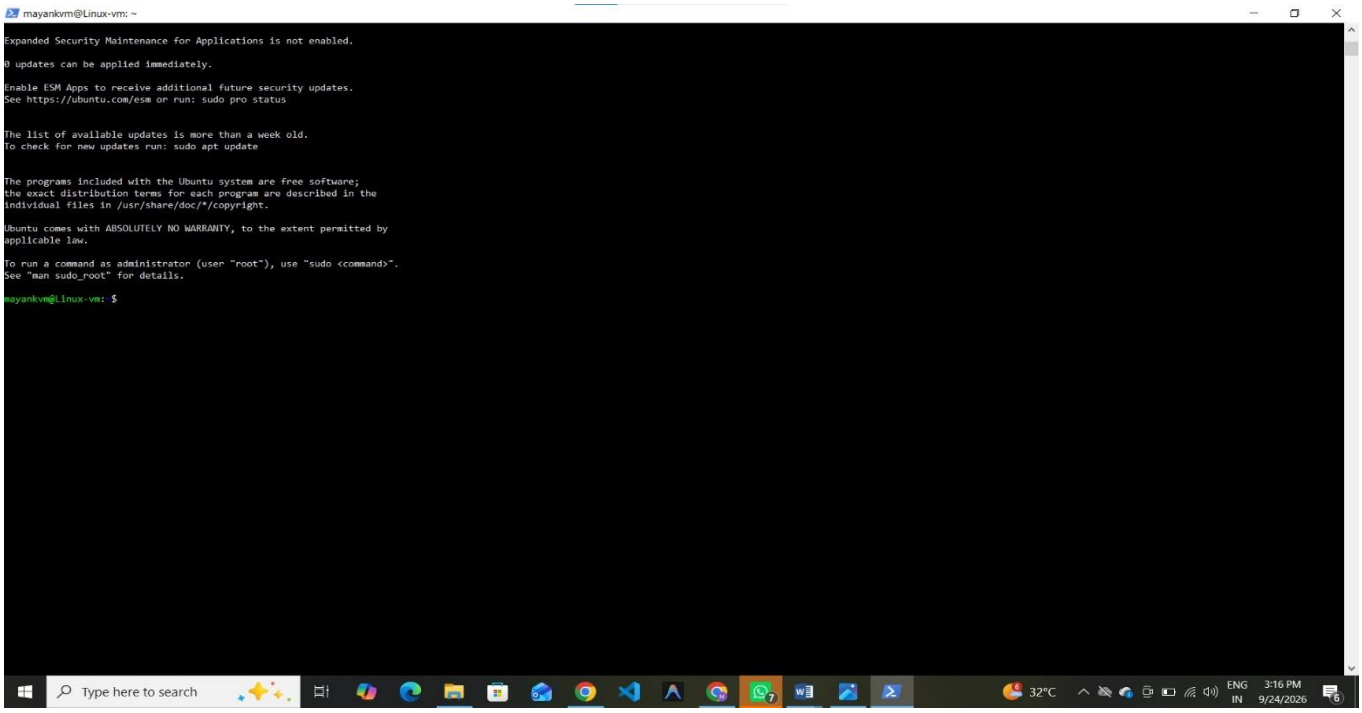
Attached Disk Details

- Disk Name: Data_Disk_1
- Storage Type: Premium SSD LRS
- Disk Size: 8 GiB
- Host Caching: Read-only



Step 5: Connect to the Linux Virtual Machine

1. Open the SSH terminal.
2. Connect to the Linux VM using the VM's Public IP Address.
3. Verify successful login.



Step 6: Verify the Newly Attached Disk

1. Execute the following command:

    - lsblk

1. Review the available block devices.
2. Verify that a new disk **sdb** appears in the output.

Observation

- `sda` represents the operating system disk.
- `sdb` represents the newly attached data disk.
- The size of the new disk is 8 GiB.

```
mayankvm@Linux-vm: ~
mayankvm@Linux-vm:~$ lsblk
NAME    MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda       8:0    0   30G  0 disk
├─sda1    8:1    0   29G  0 part /
├─sda14   8:14   0    4M  0 part
├─sda15   8:15   0  106M  0 part /boot/efi
└─sda16 259:0    0  913M  0 part /boot
sdb       8:16   0    8G  0 disk
sr0      11:0    1  628K  0 rom
mayankvm@Linux-vm:~$
```

Step 7: Format the Data Disk

1. Create an EXT4 file system on the newly attached disk using the following command:

- sudo mkfs.ext4 /dev/sdb

Purpose

Formatting prepares the disk for storing files and directories by creating a Linux-compatible file system.

```
mayankvm@Linux-vm:~$ sudo mkfs.ext4 /dev/sdb
mke2fs 1.47.0 (5-Feb-2023)
Discarding device blocks: done
Creating filesystem with 2097152 4k blocks and 524288 inodes
Filesystem UUID: 82852675-2063-4cdb-9f26-f352f63fb104
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

mayankvm@Linux-vm:~$
```

Step 8: Create a Mount Point and Mount the Disk

Create a directory to use as a mount point:

- sudo mkdir /datadisk

Mount the disk to the directory:

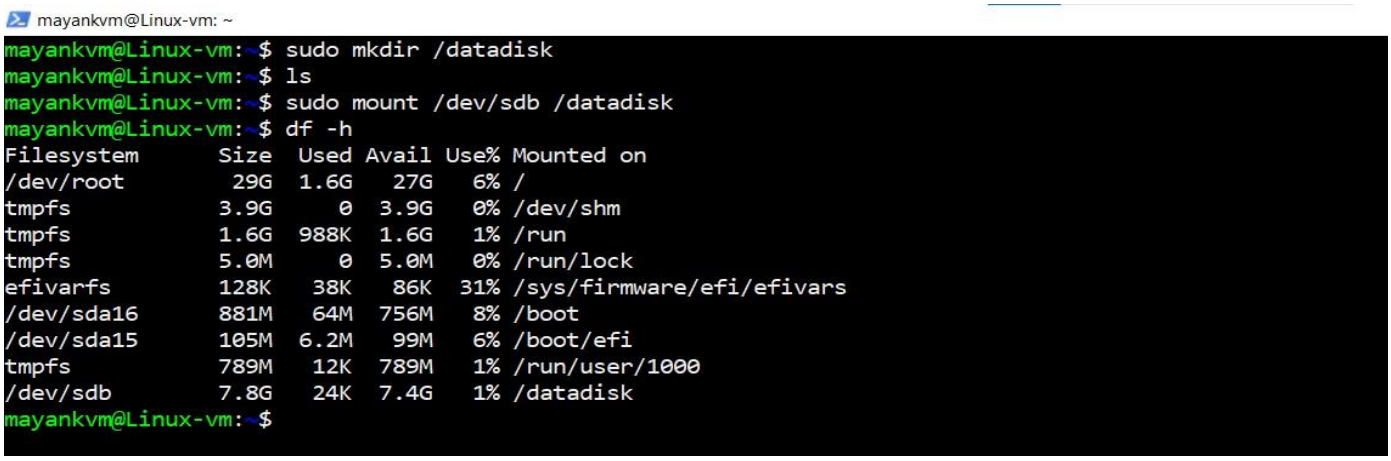
- sudo mount /dev/sdb /datadisk

Verify the mount status:

- df -h

Observation

- Disk `/dev/sdb` is mounted successfully.
- Mount Point: `/datadisk`
- Available Space: Approximately 8 GB



Step 9: Verify Read and Write Access

Navigate to the mounted directory:

- cd /datadisk

Create a test file:

- sudo touch mayank

List the files:

- ls

Verify that the file has been created successfully.

Observation

- Folder Present: `lost+found`
- Test File Created: `mayank`
- Disk is operational and writable.

```
mayankvm@Linux-vm:~$ cd /datadisk/
mayankvm@Linux-vm:/datadisk$ ls
lost+found
mayankvm@Linux-vm:/datadisk$ touch mayank
touch: cannot touch 'mayank': Permission denied
mayankvm@Linux-vm:/datadisk$ sudo touch mayank
mayankvm@Linux-vm:/datadisk$ ls
lost+found  mayank
mayankvm@Linux-vm:/datadisk$
```

Step 10: Unmount the Data Disk

Move outside the mounted directory:

- cd ..

Unmount the disk:

- sudo umount /datadisk

Verify using:

- df -h

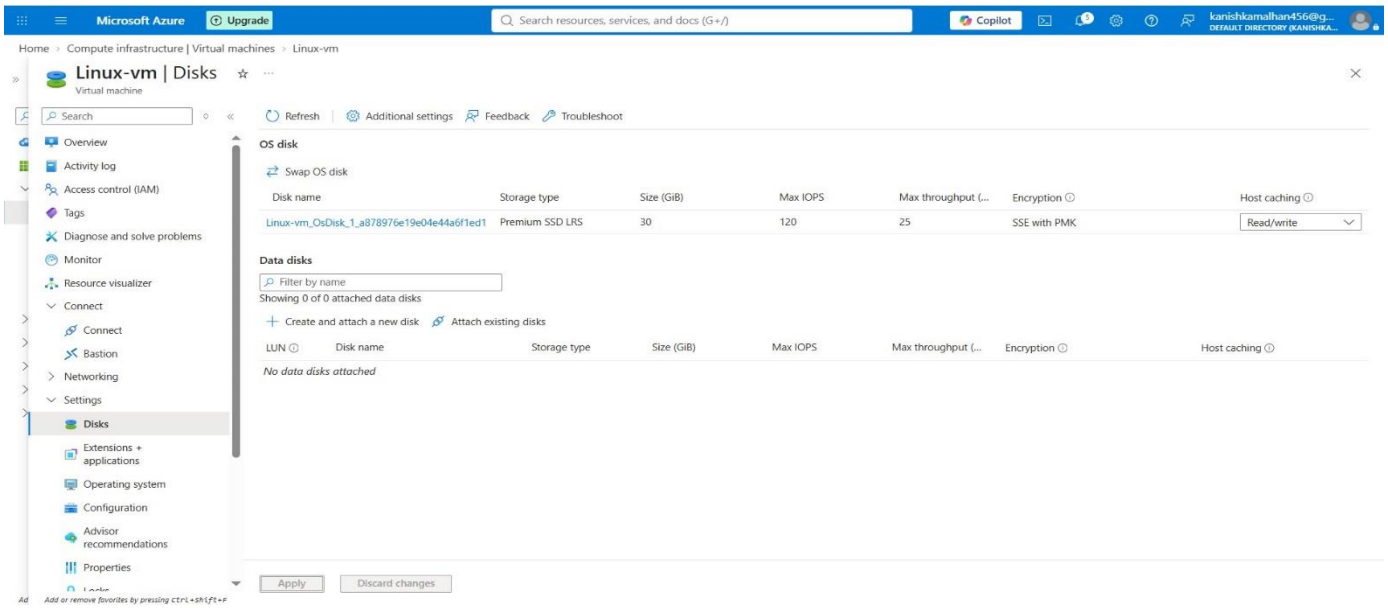
Confirm that `/datadisk` is no longer displayed.

mayankvm@Linux-vm: /

```
mayankvm@Linux-vm:/datadisk$ cd ..
mayankvm@Linux-vm:/$ sudo umount /datadisk
mayankvm@Linux-vm:/$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        29G  1.6G   27G   6% /
tmpfs           3.9G     0  3.9G   0% /dev/shm
tmpfs           1.6G  988K  1.6G   1% /run
tmpfs           5.0M     0  5.0M   0% /run/lock
efivarfs        128K   38K   86K  31% /sys/firmware/efi/efivars
/dev/sda16      881M   64M  756M   8% /boot
/dev/sda15      105M  6.2M   99M   6% /boot/efi
tmpfs           789M   12K  789M   1% /run/user/1000
mayankvm@Linux-vm:/$
```

Step 11: Detach the Data Disk

1. Return to the Azure Portal.
2. Open **VM** → **Disks**.
3. Select the attached Data Disk.
4. Click **Detach Disk**.
5. Save the configuration.
6. Refresh the page and verify that no Data Disk is attached.



Step 12: Verify Disk Removal

1. Return to the SSH session.
2. Execute:

    - lsblk

Verify that only the operating system disk remains.

Observation

- `sdb` is no longer displayed.
- The data disk has been detached successfully.
- Only OS disk `sda` remains attached.

```
mayankvm@Linux-vm: /
mayankvm@Linux-vm:/$ lsblk
NAME    MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda       8:0    0   30G  0 disk
├─sda1    8:1    0   29G  0 part /
├─sda14   8:14   0    4M  0 part
├─sda15   8:15   0  106M  0 part /boot/efi
└─sda16 259:0    0  913M  0 part /boot
sr0      11:0    1  628K  0 rom
mayankvm@Linux-vm:/$
```

Observation

A managed data disk was successfully attached to the Linux Virtual Machine, detected by the operating system, formatted using the EXT4 file system, mounted for use, and verified through file creation. The disk was then safely unmounted and detached from the VM without impacting the operating system disk.

Result

The Azure Managed Data Disk was successfully attached, verified, formatted, mounted, tested, unmounted, and detached from the Linux Virtual Machine.